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Data Article

A wearable sensors dataset for stress & boredom associated activity recognition

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ABSTRACT

This article presents a dataset of activities associated with stress and boredom obtained through wearable sensors. Data was collected from 40 right-handed participants aged 20 to 25, evenly split between males and females. Each individual wore a smart device on their dominant arm's wrists to facilitate the capture of data. This dataset covers five activities associated with stress and boredom, namely, smoking, eating, nail biting, face touching, and staying still. These activities were selected for their potential psychological implications and captured in an uncontrolled environment to mimic real-life scenarios. The data provides a unique resource for developing machine learning models aimed at recognizing these behaviors, which could lead to real-time analysis and interventions for stress. A custom holder was used to hold the device on the wrists in order to ensure that all participants had consistent orientation and placement. This holder was situated just above the wrist joint, a location typically associated with the placement of smartwatches. The dataset provides a unique opportunity for developing machine learning models

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for stress & boredom associated activities recognition apart from real-time symptomatic analysis of stress and boredom.

Specifications Table

Subject	Artificial Intelligence, Machine Learning
Specific subject area	Human activity recognition through wearable sensors
Data format	Raw data from motion sensors
Type of data	CSV file of sensors time series data
Data collection	A sensors' device, Samsung Galaxy S5, was worn on the wrist of the dominant arm by subjects. To maintain consistency in orientation and placement for all participants, a specialized holder was employed. This holder was positioned slightly above the wrist joint of the dominant hand, a location commonly used for smartwatches. The participants were then asked to perform different activities associated with stress and boredom. They were not instructed about how to perform an activity. The duration of each activity was kept around one minute with the exception of smoking activity. In that case it was variable as it was the time to finish a cigarette.
Data source location	School of Mathematics and Computer Science, IBA, Karachi, Pakistan
Data accessibility	Repository name: Mendeley repository Data identification number: 10.17632/2dn3hpbm5m.1 Direct URL to data: https://data.mendeley.com/datasets/2dn3hpbm5m/1
Related research article	Khan, Nida Saddaf, Saleeta Qadir, Gulnaz Anjum, and Nasir Uddin. "StresSense: Real-Time detection of stress-displaying behaviors." International Journal of Medical Informatics 185 (2024): 10,540, https://doi.org/10.1016/j.ijmedinf.2024.105401.

1. Value of the Data

The data serve multiple valuable purposes:

- It is instrumental in validating the application of wearable sensors for recognition of activities associated with stress and boredom.
- Researchers and developers can leverage this dataset to design novel non-intrusive systems through wearable sensor data.
- These data can function as a gold standard for assessing the performance of various algorithms used for recognition of events associated with stress and boredom.
- Furthermore, the data enables a direct comparison between data collected from wearable sensors and traditional approaches for stress associated activities analysis.

2. Data Description

The dataset was created with an intention aimed at verifying the accuracy of wearable sensor measurements in recognizing stress and boredom related activities recognition. This dataset is structured with a primary folder named 'StressBoredomData'. This folder contains the data file "StresSense.csv".

The dataset comprises a total of 495,446 data points with twelve columns. These columns include a timestamp, user ID, activity label, and three-dimensional values from the accelerometer (A), gyroscope (G), and magnetometer (M), labeled as follows [1].

- TimeStamp: time stamp of a record with sampling rate of 50 Hz.
- Gx: angular velocity on the x axis of the gyroscope.

- Gy: angular velocity on the y axis of the gyroscope.
- Gz: angular velocity on the z axis of the gyroscope.
- Ax: acceleration on the x axis of the accelerometer.
- Ay: acceleration on the y axis of the accelerometer.
- Az: acceleration on the z axis of the accelerometer.
- Mx: magnitude on the x axis of the magnetometer.
- My: magnitude on the y axis of the magnetometer.
- Mz: magnitude on the z axis of the magnetometer.
- User: unique identification number for each user.
- Activity: name of an activity being performed.

The data collection method aligns with previous approaches used for collecting anxiety-related data, which used an Inertial Measurement Unit (IMU) worn on the wrist [2]. For this data collection, the sensor devices are worn at the wrist position of the dominant arm of the participant.

There are other relevant studies where the dataset was collected for human activity recognition. The WISDM dataset includes raw and transformed data for activities like walking, jogging, and climbing stairs, captured via smartphone accelerometers [3]. Data was collected from 29 volunteers who carried Android phones in their front pants pockets. These individuals performed specified activities while the devices recorded data at a rate of 20 samples per second, aiming to enhance fitness related activity recognition algorithms. Similarly, the human activity recognition dataset at the UCI Repository involves data from 30 volunteers, performing activities of daily living (ADLs) like walking and sitting while wearing a waist-mounted smartphone [4]. This dataset includes 10,299 instances with data from the smartphone's accelerometer and gyroscope, capturing 3-axial linear acceleration and angular velocity at 50 Hz. Another dataset from UCI repository is PAMAP2 for physical activity monitoring [5]. It consists of data from 9 subjects performing 18 different physical activities, recorded via three inertial measurement units and a heart rate monitor. The dataset includes over 3.85 million instances, captured at a high sampling frequency (100 Hz for IMUs and ~9 Hz for heart rate monitor). Various datasets are available; some are publicly accessible, while others are curated specifically for particular studies [6–8]. Our dataset contributes to the research community by allowing researchers to replicate the results discussed in the associated paper [9] and utilize the data in their studies to further enrich the knowledge base. This dataset used in the study to detect the stress and boredom related activities in real-time [9]. In this study, a deep neural network (DNN) was learned along with conventional machine learning models to detect the activities in real-time. For this, the data underwent several preprocessing steps, including dealing with missing values and applying normalization techniques. Utilizing multiple sensors increased the likelihood of missing data due to different recording delays associated with each sensor type. Initial segments of the data stream that lacked values were discarded because they generally did not contribute to the analysis of the activities being monitored. In cases where missing data occurred during the activity, the gaps were filled using the average of the five nearest recorded values, either before or after the missing value. Normalization procedures were also implemented to align all data streams to a uniform scale, a critical step in preparing data for DNN training. This standardization process aims to center the data around a zero mean, which helps in speeding up the learning process and achieving faster convergence.

The data transformation techniques are necessary for employing machine learning algorithms on the time series data collected from sensors. This involved converting the continuous data into sliding windows, possibly overlapping, to enrich the training dataset. According to the reference [10], an 80% overlap in window segmentation was found to be optimal, a practice that was adopted for this study to enhance the model's learning potential.?

This research explores two methods for processing sensor data to recognize human activities: using raw sensor data and using statistical features derived from the data. In the raw data approach, sensor readings are directly input into a neural network. For example, if an accelerometer's data is used with a window size of 50, this results in 150 input neurons corresponding to the three dimensions (x, y, z) of the accelerometer. Conversely, the statistical features approach involves inputting calculated mean, standard deviation, minimum, and maximum values for each dimension into the neural network. This results in fewer neurons—24 for the same accelerometer and gyroscope setup, utilizing four statistical features for each of three dimensions.

The research utilizes a DNN model to differentiate between human activities, a common application in pattern recognition requiring large datasets. The model's accuracy is evaluated using the hold-one-out method, where data from one participant is withheld as a test set while the remainder is used for training. This DNN model is compared against traditional machine learning algorithms such as Decision Trees, Naïve Bayes, K-Nearest Neighbors, and ensemble methods like Random Forests and Gradient Boosting, which serve as benchmarks in human activity recognition (HAR). The paper also assesses the real-time performance of the best model to determine how quickly it can identify activities.

3. Experimental Design, Materials and Methods

Data collection was performed using an Android application [11] at a sampling rate of 50 Hz, as studies have indicated that this frequency captures the range of human movement [12]. Recordings were initiated and stopped manually, with each activity's data stored locally. There was a total of 40 healthy, right-handed participants (20 males and 20 females) aged 20–25, although not all participants performed every activity. Activities were conducted in an uncontrolled environment, with each participant assigned specific activities to perform.

Labeling was accomplished using the Android app [11], allowing the activity name to serve as the file name. Data for each activity was recorded sequentially, and later, all instances were assigned the corresponding activity label. The same procedure was followed for all participants. Fig. 1 provides an overview of the number of instances related to the activity and user ID. Fig. 1(A) illustrates that the number of instances for each activity is nearly equal, indicating a balanced class distribution. In contrast, Fig. 1(B) reveals variations in the number of instances recorded by each user, with some recording all five activities and others only one or two from the set. It's important to note that the data collection process was entirely voluntary and posed no risk of emotional or physical harm. Participants had the freedom to terminate their participation at any time.

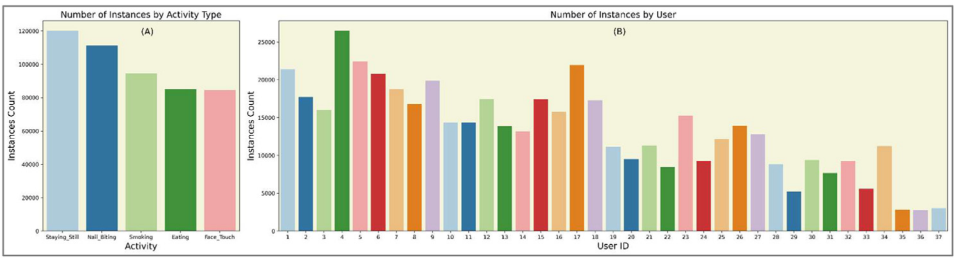

Fig. 1. (A) represents instances count by activity type, (B) represents instances count by user.

The selection of activities for our dataset focused on stress and boredom, we would like to emphasize the evidence-based rationale grounded in contemporary psychological and behavioral research [8]. Activities such as smoking are often perceived as mechanisms for stress relief and are substantiated by recent studies illustrating their use as coping strategies during stressful episodes [13,14]. This supports the inclusion of smoking in our dataset as it serves as a self-soothing behavior that individuals may engage in to manage acute stressors, making it relevant for real-time monitoring focused on stress behaviors.

Moreover, the relationship between smoking and emotional distress is complicated [15], which illustrates how the initiation of smoking is associated with increased depressive distress among participants which may overshadow some of the correlations. This highlights a complex interaction between behaviors such as smoking and mental health that extends beyond immediate relief to potentially more chronic emotional states. Such findings are critical for our dataset, which seeks to capture a range of behaviors indicative of both stress and boredom. This inclusion underscores the complex nature of smoking as both a response to immediate stress and a factor in depression, making it a valuable behavior for studies aiming to understand and intervene in these psychological states.

Our dataset is designed not only to track behaviors typically associated with stress and boredom but also to facilitate the development of applications that utilize real-time data for effective management and intervention. These applications, ranging from health monitoring systems to behavioral intervention tools, leverage insights into behaviors like smoking to enhance individual and organizational well-being. Additionally, the inclusion of varied activities, such as "staying still," often misinterpreted as a sign of calmness but hypothesized in our research as an indicator of disengagement or low mood/ depression, allows for a comprehensive analysis. This approach enriches psychophysiological research and supports the development of personalized interventions, thereby advancing our understanding of stress and boredom management across diverse settings. Through detailed analysis and sophisticated machine learning techniques, our study is an effort to explore these relationships between diverse activities and complex emotional states, enhancing the practical applications of our findings in clinical, personal, and occupational health domains.

Fig. 2 provides a visualization for all three sensors time series data against all five activities. Each subplot represents the three axis readings for a five second duration for the user with ID 1. In the following subsections, we will elaborate on the recorded activities within the context of this research.

3.1. Smoking

Smoking can be characterized as a time-dependent action. A smoking session entails at least one puff, defined as the duration it takes to bring the cigarette to one's lips, inhale, and lower their arm. Data on smoking were collected until a participant finished smoking an entire cigarette.

3.2. Eating

Eating takes on various forms globally. Measuring eating movements is complex due to (1) individual preferences in using utensils, diverse manners, and cultural dining customs, and (2) unique personal habits such as touching one's hair or face. This diversity makes it challenging to develop a universal model for all users [16]. To mitigate these complexities, a protocol was established to minimize variations in the pattern of this activity. Eating was recorded for chocolate consumption and continued until a participant finished the chocolate bar.

3.3. Nail-Biting

Onychophagia, commonly known as nail-biting, is a response to nervousness, stress, hunger, or boredom. It is both unsanitary and detrimental to one's nails [17]. In this study, habitual nail biters were instructed to engage in the activity while sitting and standing. It was observed that nail-biting primarily involved wrist movements near the face, unlike the elbow and shoulder movements observed in eating and smoking. Participants performed this activity for three minutes.

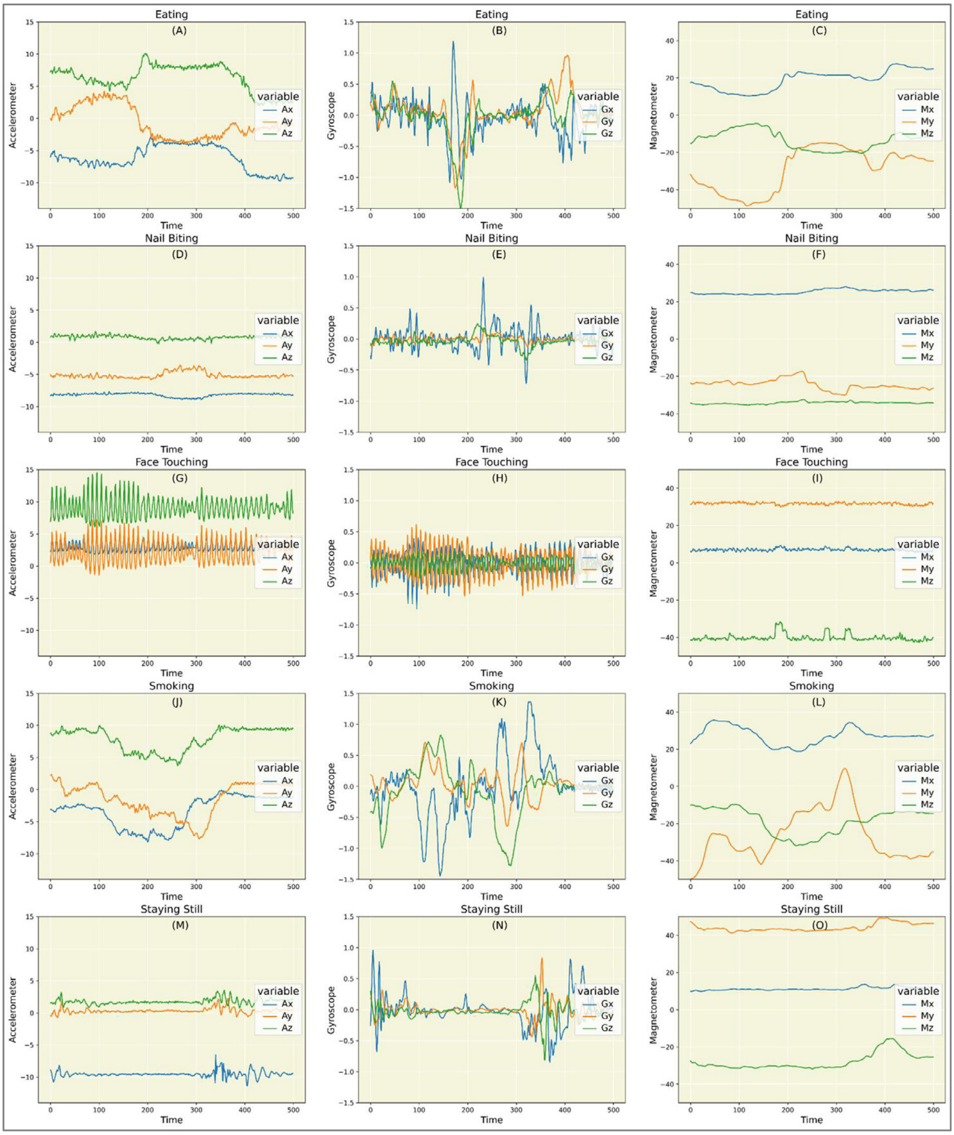

Fig. 2. (A–O) represents plots for all three sensors readings for five seconds for all five activities for user 1. Ax, Ay and Az represent *x*-axis, *y*-axis and *z*-axis readings for accelerometer, Gx, Gy and Gz represent *x*-axis, *y*-axis and *z*-axis readings for gyroscope and Mx, My and Mz represent *x*-axis, *y*-axis and *z*-axis readings for magnetometer.

3.4. Face touching

Face touching encompasses contact with any part of one's face, including the head, hair, or neck. Occasional and spontaneous touches are normal, but the frequency of face touching appears to be linked to negative emotions and distraction of attention. This behavior may play a role in emotion regulation [18,19]. The protocol specified areas of the face following Centers for Disease Control and Prevention (CDC) guidelines (eyes, nose, and mouth) and other commonly touched regions (hair, forehead, temple, ear, cheek, and chin) [20]. Participants were instructed to touch their faces as they typically would in both sitting and standing positions, and this activity was carried out for three minutes.

3.5. Staying still

Remaining still served as the baseline condition; participants were asked to remain at rest and refrain from engaging in other activities. The protocol allowed for folding arms, placing hands on hips, and fidgeting with fingers. Some participants were observed swaying or fidgeting, while others remained completely motionless. This activity was performed in both standing and sitting positions for a duration of three minutes.

Limitations

While the presented dataset offers valuable insights into activities related to stress and boredom, it comes with certain limitations. The sample size of 40 individuals, comprising 20 males and 20 females aged 20 to 25, may not fully capture the diverse population. Additionally, the dataset covers a small range of daily activities associated with stress and boredom, and not all participants engaged in every activity, potentially limiting the scope. Moreover, the use of a custom holder for consistent sensor placement could influence the natural behavior of participants. Researchers should be mindful of these limitations when utilizing the dataset for stress and boredom-associated activity recognition or real-time symptomatic analysis.

Ethics Statement

Ethical approval for our study was granted by the Capital University of Science and Technology (CUST), Pakistan, with the Ethical Review Report number: CUST/ORIC/IERB/2023/02. Adhering to ethical standards in Pakistan, verbal consent was obtained from all participants, considering logistical challenges and cultural barriers in obtaining written consent in certain regions. The process involved thorough and respectful communication, ensuring participants were fully informed about the study's nature, purpose, and their right to withdraw without consequences. Participant anonymity was meticulously documented, upholding the integrity and ethical rigor of our research approach.

Data Availability

A Wearable Sensors Dataset for Stress & Boredom Associated Activity Recognition (Original data) (Mendeley Data).

CRediT Author Statement

Saleeta Qadir: Writing – original draft; **Nida Saddaf Khan:** Conceptualization, Methodology, Data curation, Supervision; **Gulnaz Anjum:** Writing – review & editing; **Nasir Uddin:** Writing – review & editing.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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